Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/07/2018
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	GCR Simulator Validation Studies with Human and Mouse Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	a_kronenberg@lbl.gov	**Fax:**	FY 510-486-4475
PI Organization Type:	GOVERNMENT	**Phone:**	510-486-6449
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Road, Building 977-180		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	5
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2019 per PI and approval by NASA LaRC (Ed., 12/19/18)		
Key Personnel Changes/Previous PI:	No key personnel changes.		
COI Name (Institution):	Turker, Mitchell Ph.D. (Oregon Health & Science University) Grossi, Gianfranco Ph.D. (Universita Federico II) Guida, Peter Ph.D. (Brookhaven National Laboratory) Grygoryev, Dmytro Ph.D. (Oregon Health and Science University) Rwatambuga, Furaha B.S. (Oregon Health and Science University) Bonti, Kristi A.A. (Brookhaven National Laboratory) Gauny, Stacey M.S. (Lawrence Berkeley National Laboratory) Raber, Jacob Ph.D. (Oregon Health & Science University) Torres, Eileen Ruth S. B.S. (Oregon Health & Science University) Akinyeke, Tunde Ph.D. (Oregon Health & Science University) Lee, Joanne (Oregon Health & Science University) Herinckx, Molly (Oregon State University) Cisneros, Miguel Legaria (Oregon Health & Science University)		

Grant/Contract No.:	NNJ16HP23I
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This project addresses NASA's request to help design and validate its planned galactic cosmic radiation (GCR) simulator. Space radiation has two major components: GCR, which is comprised of highly energetic ions from protons through uranium, and solar particle events, which are primarily moderate energy protons. Most currently funded NASA ground-based studies in radiobiology examine one highly energetic ion at a time because of facility limitations. While this approach provides insight into the biological effect of each particle, insight into potential interactions from particles of different energies or charges is missed and the deep space environment is not accurately copied. Recent improvements in facility capability allow mixed beam exposures to more closely approximate the GCR, but questions remain about the final composition of the beams to be included. We will build upon our prior work on single particle-induced mutations in cultured cells and mice (cancer-relevant endpoints). Work with cultured cells will be performed because it will provide data more quickly, aid with cross-species extrapolation of risks, and inform the mouse work. We will first measure the effects of acute exposures to alpha particles (charged helium nuclei) because basic information about this GCR component is lacking. Next, we will measure the effect of acute exposures to a mixture of protons, alpha particles, and silicon ions to mimic the effect of a simple GCR simulator model, followed by acute exposures to the final GCR mixture that is chosen (informed in part from our first two experiments). Finally, we will determine the effects of a protracted exposure to the GCR simulator, which will reflect more closely exposures that will occur in the space environment. Our results will help NASA to both design and validate the GCR simulator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of this project is to provide input to NASA's risk model for cancer regarding exposure to accelerated charged particle radiations that simulate those found in space. The work is designed to address cancer development processes at relatively low doses and fluences that are space-relevant. The project specifically uses different types of charged particle radiations alone or in combination to determine whether it is possible to predict the effects of multiple ions from the effects measured in studies using single light or heavy charged particle beams. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.
Task Progress:	Task progress in the second year included completion of studies for He ion toxicity (250 MeV/n, LET = 1.6 keV/µm) in human B-lymphoblasts (Ed. note: LET is linear energy transfer). On a per unit dose basis, the results of replicate experiments could not be distinguished from the results obtained previously for the same cell line exposed to 250 MeV protons (LET = 0.4 keV/µm) or 55 MeV protons (LET = 1.46 keV/µm). On a per particle basis, the 250 MeV/n He ions have a similar response to the 55 MeV protons, and a four-fold increase in toxicity compared with 250 MeV protons. Studies on the cytogenetic effects of He ions (250 MeV/n) were also conducted in B6D2F1 kidney epithelial cells exposed in situ and monitored several months after exposure. Dose-dependent increases in chromosome type and chromatid type aberrations were similar if all cells with aberrations were considered. If cells with two or more aberrations were considered, chromosome type aberrations were predominant. Studies on behavior and cognition were also performed in mice exposed to He ions. Studies with three rapidly delivered sequential ion beams were conducted using cultured kidney epithelial cells from B6D2F1 mice. Cytotoxicity and autosomal mutation were assessed using standard methods. The results of replicate experiments indicate dose-dependent cell killing and mutation induction that is being compared with the results from the individual ion beams obtained previously. Studies were also performed using a cohort of B6D2F1 mice to monitor cytogenetic changes, cytotoxicity and mutation exposed to very similar rapidly delivered sequential ion beams. The results are pending. Supplemental studies on behavior and cognition were also performed.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A. "Chromatin remodeling." Symposium S13 - 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. Abstract Book. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kronenberg A, Grygoryev D, Gauny S, Snijders AM, Mao J-H, Lasarev M, Ohlrich A, Rwatambuga F, Johnson S, Eckelmann B, Hryciw G, Turker M. "Simulated space radiation-induced mutants in the mouse kidney display widespread genomic change." Poster Session 5 - 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. Abstract book. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Grossi G, Lee J, Weber S, Grygoryev D, Turker M, Raber J. "GCR simulator validation studies with human and mouse models." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Articles in Peer-reviewed Journals	Turker MS, Grygoryev D, Lasarev M, Ohlrich A, Rwatambuga FA, Johnson S, Dan C, Eckelmann B, Hryciw G, Mao JH, Snijders AM, Gauny S, Kronenberg A. "Simulated space radiation-induced mutants in the mouse kidney display widespread genomic change." PLoS One. 2017 Jul 6;12(7):e0180412. eCollection 2017. https://doi.org/10.1371/journal.pone.0180412 ; PubMed PMID: 28683078; PubMed Central PMCID: PMC5500326 , Jul-2017